

1. An urn contains 4 red and 8 blue balls. Five of the balls are chosen from the urn, one at a time, and placed in a row on the table. What is the probability that:
 - a. the first three balls chosen are red, and then the next two are blue;
 - b. the first two balls chosen are blue, and then the next three are red;
 - c. the first ball is red, the second blue, the third red, the fourth blue, and the fifth red;
 - d. the first two balls are red, the third and fourth balls are blue, and the fifth ball is red;
 - e. the first and fifth balls are blue, and the second, third, and fourth balls are red?
2. An urn contains 4 red and 8 blue balls. Five of the balls are chosen from the urn, one at a time, and placed in a row on the table. What is the probability that three red, and two blue balls are chosen (in any order).
3. A bag contains 4 red and 8 blue balls. Five of the balls are taken from the bag (at the same time). What is the probability that three of the chosen balls are red, and the other two are blue?
4. A playlist contains 14 songs by Drake, and 8 by Kendrick Lamar (and no other songs). When the playlist is played on 'shuffle' the songs are played in a random order without repeats. If I listen to six songs, what is the probability that:
 - a. the first three songs are by Drake, and the next three are by Kendrick;
 - b. the songs alternate Drake, Kendrick, Drake, Kendrick, Drake, Kendrick;
 - c. the songs alternate Kendrick, Drake, Kendrick, Drake, Kendrick, Drake;
 - d. the first two songs are Drake, the next two are Kendrick, the fifth song is Drake and the sixth is Kendrick;
 - e. the first, fourth, and fifth songs are Drake, and the remaining songs are Kendrick?
5. A playlist contains 14 songs by Drake, and 8 by Kendrick Lamar (and no other songs). When the playlist is played on 'shuffle' the songs are played in a random order without repeats. If I listen to six songs, what is the probability that I hear an equal number of Drake and Kendrick songs (in any order).
6. A bag contains 5 red, and 3 blue balls. If three balls are chosen from the bag then what is the probability that two blue balls and one red ball are chosen?
7. An office contains 9 women and 5 men. The board of directors create a leadership team at random from four members of the office. What is the probability that the team consists of:
 - a. all women;
 - b. two women and two men;
 - c. at least one man?

In fact, the leadership team is composed entirely of men. Comment.
8. One urn contains three blue balls and seven red balls. Another urn (identical in appearance to the first) contains three blue balls and three red balls. You friend choses one urn at random, and pulls out three balls.
 - a. What is the probability that they pull out two red balls and one blue ball?
 - b. Given that they pull out two reds and one blue ball, what is the probability that they chose the first urn (with seven red balls originally inside)
9. A fair die is rolled ten times. What is the probability that exactly three sixes are rolled?
10. A commuter models the probability that his bus to work will be late with probability $\frac{1}{3}$, independently of whether or not the bus was on late on any other day. If they commuter travels to work five days a week, then what is the probability that his bus will be late on:
 - a. all five days;
 - b. exactly two days;
 - c. at least two days.